

PLAIN LANGUAGE GUIDE

Pharmacovigilance Made Simple.

The story of Martin, a medicine called Relvian, and how the global drug safety system works — told in plain language for anyone who wants to understand pharmacovigilance.

● 9 Chapters ● Plain Language ● Free Download



What is pharmacovigilance?

REGULATORY DEFINITION

"The science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other medicine-related problem."

Article 1(28d), Directive 2001/83/EC as amended by Directive 2010/84/EU

Once a medicine is authorised and reaches patients, scientific scrutiny does not cease. Real-world patient populations differ substantially from clinical trial cohorts: they are older, carry multiple comorbidities, and are frequently prescribed concurrent therapies.

Some adverse effects are too infrequent to manifest during pre-authorisation studies; others emerge only after prolonged exposure or in specific patient subgroups.

Pharmacovigilance is the regulatory and scientific framework that identifies, evaluates, and responds to these risks throughout a medicine's entire authorised lifecycle.

It depends on healthcare professionals, patients, manufacturers, and regulatory authorities each fulfilling a defined and legally mandated role.

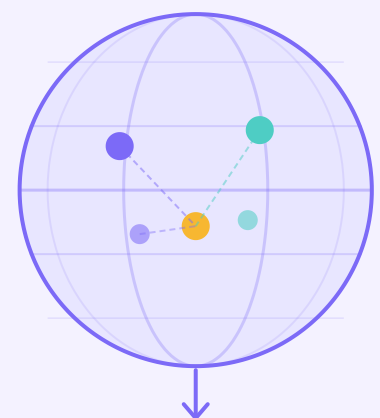
130+

countries in the WHO
International Drug Monitoring
Programme

30M+

case reports in VigiBase, the
global safety database

"Pharmacovigilance is the reason a side effect discovered in one country can protect a patient on the other side of the world."



VIGIBASE

Global Safety Database
WHO, Uppsala Monitoring Centre

- EU / EMA
- WHO Members
- Signal origin

This is Martin.

Martin is 61 years old and has osteoarthritis in his knees. His pain has gradually worsened, and it is no longer well controlled with exercise and occasional over-the-counter pain relief.

REGULATORY CONTEXT

"Patient" means any individual who uses or receives healthcare. Patients are recognised as key stakeholders in pharmacovigilance reporting under GVP Module VI (Rev 2).

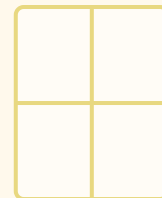
EMA GVP Module VI Rev 2, 2017

HIS DIAGNOSIS

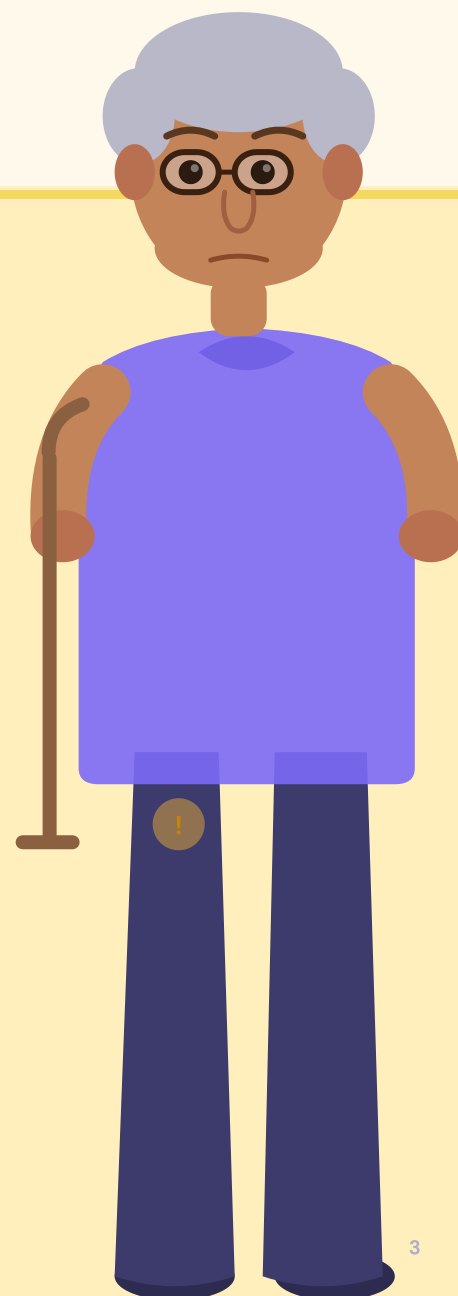
Osteoarthritis is a degenerative joint disease characterised by cartilage breakdown, progressive pain, and reduced mobility. It affects more than 500 million people worldwide and is among the leading causes of chronic pain in adults over 50. For Martin, the condition is worsening to the point where standard analgesics are no longer sufficient.

WHY MARTIN'S STORY MATTERS

Martin represents one of millions of patients prescribed newly authorised medicines every year. His experience — and how it was handled — illustrates the pharmacovigilance system working exactly as it was designed to. Follow his journey through each of the nine chapters that follow.



MARTIN
Age 61 · Osteoarthritis



Medical History and Treatment Initiation.

Martin's doctor prescribes him Relvian, a new anti-inflammatory medicine that has just come on the market. Martin collects the medicine from his local pharmacy. The pharmacist informs him of the known potential side effects: indigestion, dizziness, and mild headache.

After a week, Martin develops unusual shortness of breath when climbing stairs and a persistent dry cough. He also feels more tired than usual. He wonders if it has something to do with his new medicine. He makes an appointment with his doctor to find out.

REGULATORY CONTEXT

"Member States shall take all appropriate measures to encourage patients, doctors, pharmacists and other healthcare professionals to report suspected adverse reactions to the competent authority."

Article 107a, Directive 2001/83/EC

SIDE EFFECTS KNOWN AT TIME OF PRESCRIPTION

Indigestion · Dizziness · Mild headache.
Drug-induced lung inflammation was **not** listed as a known adverse effect of Relvian at the time of prescription.

WARNING SIGNS (ONE WEEK LATER)

Unusual shortness of breath when climbing stairs · Persistent dry cough · Increased fatigue. Martin suspects a link to his new medicine and contacts his doctor.

PRESCRIPTION			Issued by attending physician
PATIENT Martin · Male · 61 years	SUSPECT MEDICINE Relvian · 50 mg	DOSAGE Once daily with food	
INDICATION Osteoarthritis · Knee pain	DURATION Ongoing · Review in 4 weeks	ADVERSE EVENT TRIGGERED Drug-induced pneumonitis	

Clinical Evaluation.

Martin's doctor examines him and orders tests, including chest imaging and blood work. The results show signs of drug-induced lung inflammation (pneumonitis). He checks that Martin is not taking any other medications that could be the cause. He also orders more tests to rule out infection or underlying lung disease. He advises Martin to stop taking Relvian.

Over the next two weeks, Martin's breathing gradually improves, and the cough disappears. His oxygen levels return to normal.

REGULATORY DEFINITION

"Serious adverse reaction means an adverse reaction which results in death, is life-threatening, requires inpatient hospitalisation or prolongation of existing hospitalisation, results in persistent or significant disability or incapacity..."

Article 1(12), Directive 2001/83/EC

DRUG-INDUCED PNEUMONITIS

Inflammation of lung tissue caused by a pharmaceutical agent. In Martin's case, the temporal association, resolution of symptoms after discontinuation, and absence of alternative aetiologies pointed to Relvian as the probable cause.

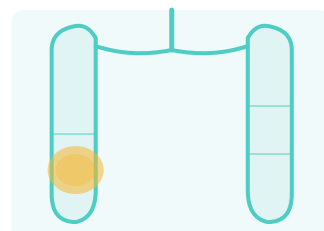
CAUSALITY ASSESSMENT

Temporal link
Present

De-challenge
Positive

Assessment
Probable

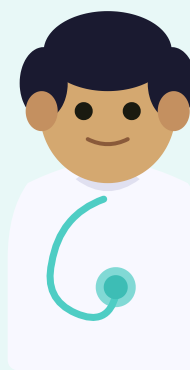
CHEST IMAGING RESULT



Inflammation detected (left lower)

FINDING

Drug-induced pneumonitis —
probable causal link to Relvian



Attending Physician

Individual Case Safety Report (ICSR) Submission.

Martin's doctor fills out a report for the adverse event Martin has experienced, as he suspects Relvian may be the cause. Drug-induced pneumonitis was not listed as a known side effect. He sends the report to the national regulatory authority.

REGULATORY DEFINITION

"Individual case safety report (ICSR) means a format for the reporting of one or more suspected adverse reactions to a medicinal product that occur in a single patient at a specific point of time."

Article 1(16a), Directive 2001/83/EC

WHAT THE ICSR CAPTURES

Patient demographics, the suspect medicine with dose and duration, a detailed narrative of the adverse reaction, the outcome, and the reporter's causality assessment. This structured format enables global databases to analyse and compare cases across countries.

INDIVIDUAL CASE SAFETY REPORT (ICSR)

Submitted to National Regulatory Authority

PATIENT Martin · Male · 61	SUSPECT MEDICINE Relvian 50 mg	DURATION OF USE 7 days · Once daily	INDICATION Osteoarthritis
ADVERSE REACTION Drug-induced pneumonitis	SERIOUSNESS Medically significant	OUTCOME Recovered	CAUSALITY Suspected

NARRATIVE SUMMARY

Male patient, aged 61, prescribed Relvian 50 mg once daily for osteoarthritis. After 7 days of treatment, developed shortness of breath on exertion and persistent dry cough. Chest imaging confirmed drug-induced pneumonitis. Relvian discontinued; symptoms resolved within two weeks. Other medications and infectious causes excluded. Reporter assessment: probable causal relationship.

Reported by: Attending Physician

Submitted to: National Regulatory Authority

Status: Transmitted to VigiBase

Safety Database Entry and Signal Detection.

REGULATORY DEFINITION

"Signal means information arising from one or multiple sources, including observations and experiments, which suggests a new potentially causal association, or a new aspect of a known association, between an intervention and an event or set of related events."

GVP Module IX Rev 1, EMA, 2017

Martin's case is added to the national database of adverse event reports for medicines and vaccines. Since the regulatory authority in Martin's country is a member of the WHO Programme for International Drug Monitoring, his report is shared with the global database, VigiBase.

At the regulatory authority, Martin's report is reviewed by a case assessor who finds a small number of similar reports involving Relvian and lung inflammation in both the national database and VigiBase. Further analysis suggests it is probable that Relvian may be associated with drug-induced pneumonitis.

SIGNAL WORKFLOW

CASE RECEIVED

Martin's ICSR entered into national adverse event database.

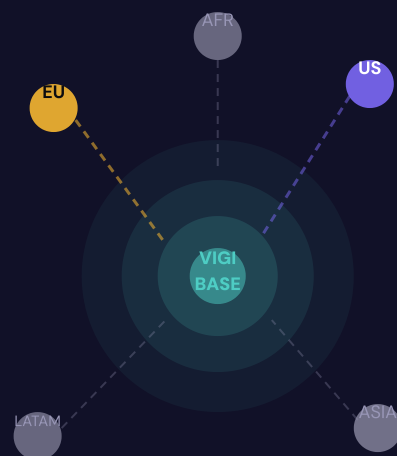
SHARED GLOBALLY

Report transmitted to VigiBase (WHO Uppsala Monitoring Centre).

PATTERN IDENTIFIED

Case assessor finds cluster of similar Relvian lung inflammation reports.

SIGNAL CONFIRMED



Probable causal association assessed. Signal raised for regulatory action.

Regulatory Action, Label Update and Risk Communication.

The regulatory authority decides that Relvian's manufacturer must update the product information. "Inflammation of the lungs" is added to the list of adverse effects, and a warning is included about the potential risk of serious respiratory reactions.

An advisory letter is sent to health providers, with additional information made available on the regulatory authority's website. Martin's doctor is informed and reviews prescribing guidance.

REGULATORY CONTEXT

"Risk minimisation measures are public health interventions intended to prevent or reduce the occurrence of adverse reactions associated with the exposure to a medicine."

GVP Module XVI Rev 2, EMA, 2017

LABEL UPDATE

"Inflammation of the lungs" added to Relvian's Summary of Product Characteristics.

WARNING SECTION UPDATED

Risk of serious respiratory reactions added to special warnings and precautions.

ADVISORY LETTER ISSUED

Direct healthcare professional communication sent. Published on regulatory authority's website.

PRESCRIBERS INFORMED

Martin's doctor reviews updated prescribing guidance for all patients on Relvian.

SUMMARY OF PRODUCT CHARACTERISTICS — BEFORE

4.8 UNDESIRABLE EFFECTS

Common (1 to 10 in 100 patients):

- Indigestion
- Dizziness
- Mild headache

Respiratory disorders: not listed

SUMMARY OF PRODUCT CHARACTERISTICS — AFTER UPDATE

UPDATED

4.8 UNDESIRABLE EFFECTS

Common (1 to 10 in 100 patients):

- Indigestion
- Dizziness
- Mild headache

NEW — Respiratory disorders

- Inflammation of the lungs (pneumonitis)

NEW — Special warning added

Risk of serious respiratory reactions.
Discontinue if lung symptoms develop.

Pharmacovigilance Outcome and Risk Minimisation.

Martin's case contributes to a better understanding of Relvian's safety profile. Reports like this help identify rare risks and improve safe use of new medicines.

Martin's doctor switches him to a different osteoarthritis treatment — pain management with physical therapy support and an alternative analgesic — and Martin remains stable with no further side effects.

"Martin did not do anything extraordinary. His doctor did not do anything extraordinary. The pharmacovigilance system worked exactly as it was designed to."

WHY THIS MATTERS

Clinical trials are conducted in controlled conditions with selected patient populations. Post-authorisation pharmacovigilance identifies signals that cannot be detected before approval. Reports like Martin's directly fulfil the legal obligation to protect public health throughout a medicine's authorised lifecycle.

MARTIN'S OUTCOME

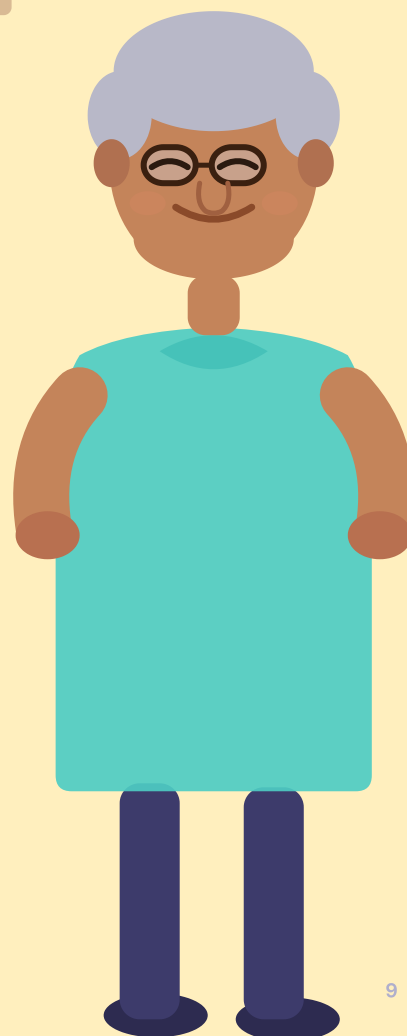
Switched to physical therapy and alternative analgesic. Stable with no further adverse effects.

SYSTEM OUTCOME

Relvian label updated, warning added, advisory letter issued to all prescribers worldwide.

STABLE

No further side effects



Glossary.

All terms from Martin's story, defined with their regulatory source.

PHARMACOVIGILANCE

The science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other medicine-related problem. (Art. 1(28d), Dir. 2001/83/EC)

ADVERSE REACTION

A response to a medicinal product which is noxious and unintended. Implies a causal relationship. Distinct from "adverse event," which does not imply causality. (Art. 1(11), Dir. 2001/83/EC)

ICSR

A format for reporting one or more suspected adverse reactions to a medicinal product that occur in a single patient at a specific point in time. (Art. 1(16a), Dir. 2001/83/EC)

SIGNAL

Information suggesting a new potentially causal association between a medicine and an adverse event, warranting further investigation. Not confirmation of causality. (GVP Module IX Rev 1)

VIGIBASE

The WHO global database of individual case safety reports, managed by the Uppsala Monitoring Centre. Contains over 30 million reports from more than 130 member countries.

RISK MINIMISATION

Public health interventions intended to prevent or reduce the occurrence of adverse reactions, or to reduce their severity or impact on patients. (GVP Module XVI Rev 2)

SUMMARY OF PRODUCT CHARACTERISTICS

The official document approved by a regulatory authority forming the basis for prescribing information. Contains approved indications, dosage, and all known adverse effects.

DRUG-INDUCED PNEUMONITIS

Inflammation of lung parenchyma caused by a pharmaceutical agent. Presents as cough, dyspnoea, and hypoxia. Diagnosis requires exclusion of infection and other aetiologies.



Martin's report made medicines safer for everyone.

He did not do anything extraordinary. His doctor did not do anything extraordinary. The pharmacovigilance system worked exactly as it was designed to. Every report matters. Every patient counts.

Qualivio supports pharmacovigilance professionals and organisations with education, consulting, and practical tools built around current regulatory frameworks.

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